# Gregory Macfarlane
## Curriculum vitae

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gregmacfarlane

## Post-graduate employment

**Brigham Young University**

since 08/2026 Associate Professor, *Civil and Construction Engineering*

11/2018-09/2026 Assistant Professor, *Civil and Construction Engineering*

**Transport Foundry**

05/2017-11/2017 Transportation Engineer

**University of North Carolina, Chapel Hill**

01/2017-05/2017 Adjunct Lecturer, *City and Regional Planning*

**WSP | Parsons Brinckerhoff**

10/2016-05/2017 Technical Principal, *Systems Analysis Group*

05/2014-10/2016 Traffic Engineer, *Systems Analysis Group*

## Higher education

**Georgia Institute of Technology**

2014 PhD, *Civil Engineering*

2014 MS, *Economics*

**Brigham Young University**

2009 BS, *Civil Engineering*, with University Honors

## Publications

† BYU undergraduate student ‡ BYU graduate student

### Refereed journal articles

[21] ‡Gibbons, N. & **Macfarlane, G. S.** (2025). Evaluating the Impacts of Parameter Uncertainty in a Practical Transportation Demand Model. *Future Transportation*, *5*(1), 15. https://doi.org/10.3390/futuretransp5010015

[20] **Macfarlane, G. S.**, ‡Barnes, M. & †Gibbons, N. (2025). A Utility-Based Approach to Modeling Systemic Resilience of Highway Networks with an Application in Utah. *Journal of Transportation Engineering, Part A: Systems*, *151*(1), 04024094. https://doi.org/10.1061/JTEPBS.TEENG-8534

[19] Voulgaris, C. T., Jensen, A. F. & **Macfarlane, G. S.** (2025). You'll Never Walk Alone: Parental active escorting as an alternative to car travel to school. *Journal of Transport & Health*, *44*, 102147. https://doi.org/10.1016/j.jth.2025.102147

[18] **Macfarlane, G. S.**, ‡Riches, G., ‡Youngs, E. K. & Nielsen, J. A. (2024). Classifying Location Points as Daily Activities using Simultaneously Optimized DBSCAN-TE Parameters.. *Findings*. https://doi.org/10.32866/001c.116197

[17] Voulgaris, C. T., **Macfarlane, G. S.** & Kaylor, J. C. (2024). Whose Miles Are These Anyway?: Estimating Site-Generated Vehicle Miles Traveled. *Journal of the American Planning Association*, *90*(4), 593–609. https://doi.org/10.1080/01944363.2023.2298962

[16] ‡Wang, B., Fulda, N., †Huang, Z., Schultz, G. G., **Macfarlane, G. S.**, †Arnesen, J. & †Khayyat, A. (2024). Predicting Directional Traffic Volume at Intersections with Automated Traffic Signal Performance Measures Data Using Machine Learning Algorithms. *Transportation Research Record*, 03611981241252829. https://doi.org/10.1177/03611981241252829

[15] ‡Wang, B., Schultz, G. G., **Macfarlane, G. S.**, Eggett, D. L. & †Davis, M. C. (2023). A Methodology to Detect Traffic Data Anomalies in Automated Traffic Signal Performance Measures. *Future Transportation*, *3*(4), 1175–1194. https://doi.org/10.3390/futuretransp3040064

[14] ‡Daines, T. J., Schultz, G. G. & **Macfarlane, G. S.** (2022). Evaluating Real Time Ramp Meter Queue Length and Wait Time Estimation. *Future Transportation*, *2*(4), 807–827. https://doi.org/10.3390/futuretransp2040045

[13] **Macfarlane, G. S.**, Voulgaris, C. T. & Tapia, T. (2022). City parks and slow streets: A utility-based access and equity analysis. *Journal of Transport and Land Use*, *15*(1), 587–612. https://doi.org/10.5198/jtlu.2022.2009

[12] **Macfarlane, G. S.**, ‡Stucki, E., Redelfs, A. H. & Spruance, L. A. (2022). Beyond Proximity: Utility-Based Access from Location-Based Services Data. *International Journal of Environmental Research and Public Health*, *19*(19), 12352. https://doi.org/10.3390/ijerph191912352

[11] ‡Wang, B., Schultz, G. G., **Macfarlane, G. S.** & †McCuen, S. (2022). Evaluating Signal Systems Using Automated Traffic Signal Performance Measures. *Future Transportation*, *2*(3), 659–674. https://doi.org/10.3390/futuretransp2030036

[10] **Macfarlane, G. S.**, ‡Sheffield, M. H., ‡Bennett, L. S. & Schultz, G. G. (2021). The Effect of Transit Signal Priority on Bus Rapid Transit Headway Adherence. *Findings*. https://doi.org/10.32866/001c.24499

[9] **Macfarlane, G. S.**, Boyd, N., Taylor, J. E. & Watkins, K. (2021). Modeling the impacts of park access on health outcomes: A utility-based accessibility approach. *Environment and Planning B: Urban Analytics and City Science*, *48*(8), 2289–2306. https://doi.org/10.1177/2399808320974027

[8] **Macfarlane, G. S.**, †Hunter, C., †Martinez, A. & †Smith, E. (2021). Rider Perceptions of an On-Demand Microtransit Service in Salt Lake County, Utah. *Smart Cities*, *4*(2), 717–727. https://doi.org/10.3390/smartcities4020036

[7] Glenn, J., †Bluth, M., †Christianson, M., †Pressley, J., Taylor, A., **Macfarlane, G. S.** & Chaney, R. A. (2020). Considering the Potential Health Impacts of Electric Scooters: An Analysis of User Reported Behaviors in Provo, Utah. *International Journal of Environmental Research and Public Health*, *17*(17), 6344. https://doi.org/10.3390/ijerph17176344

[6] Brakewood, C., **Macfarlane, G. S.** & Watkins, K. (2015). The impact of real-time information on bus ridership in New York City. *Transportation Research Part C: Emerging Technologies*, *53*, 59–75. https://doi.org/10.1016/j.trc.2015.01.021

[5] **Macfarlane, G. S.**, Garrow, L. A. & Moreno-Cruz, J. (2015). Do Atlanta residents value MARTA? Selecting an autoregressive model to recover willingness to pay. *Transportation Research Part A: Policy and Practice*, *78*, 214–230. https://doi.org/10.1016/j.tra.2015.05.010

[4] **Macfarlane, G. S.**, Garrow, L. A. & Mokhtarian, P. L. (2015). The influences of past and present residential locations on vehicle ownership decisions. *Transportation Research Part A: Policy and Practice*, *74*, 186–200. https://doi.org/10.1016/j.tra.2015.01.005

[3] Binder, S., **Macfarlane, G. S.**, Garrow, L. A. & Bierlaire, M. (2014). Associations among household characteristics, vehicle characteristics and emissions failures: An application of targeted marketing data. *Transportation Research Part A: Policy and Practice*, *59*, 122–133. https://doi.org/10.1016/j.tra.2013.11.005

[2] Wall, T. A., **Macfarlane, G. S.** & Watkins, K. E. (2014). Exploring the Use of Egocentric Online Social Network Data to Characterize Individual Air Travel Behavior. *Transportation Research Record*, *2400*(1), 78–86. https://doi.org/10.3141/2400-09

[1] Mcbride, J. H., Keach, R. W., **Macfarlane, R. T.**, De Simone, G. F., Scarpati, C., Johnson, D. J., Yaede, J. R., **Macfarlane, G. S.** & Weight, R. W. R. (2009). Subsurface Visualization Using Ground-Penetrating Radar for Archaeological Site Preparation on the Northern Slope of Somma-Vesuvius: A Roman Site, Pollena Trocchia, Italy.. https://www.iris.unina.it/handle/11588/353053

## Peer-reviewed conference papers

[8] **Macfarlane, G. S.**, ‡Day, C., Erhardt, G. D. & Watkins, K. E. (2026). Identifying ridehail riders through pairing ActivitySim and BEAM. In *Procedia Computer Science* (pp. 286–293). https://doi.org/10.1016/j.procs.2026.04.038

[7] ‡Atchley, H., †Mansfield, K. W. & **Macfarlane, G. S.** (2025). A Comparative Illustration of Trip- and Activity-Based Modeling Techniques. In *International Conference on Transportation and Development 2025* (pp. 12–23). American Society of Civil Engineers. https://doi.org/10.1061/9780784486207.002

[6] Schultz, G. G., ‡Hyer, J. C. & **Macfarlane, G. S.** (2025). An Analysis of UDOT's Expanded Incident Management Team Program. In *International Conference on Transportation and Development* (pp. 431–443). American Society of Civil Engineers. https://doi.org/10.1061/9780784486207.037

[5] ‡Jarvis, D. L., **Macfarlane, G. S.**, †Woolley, B. & Schultz, G. G. (2024). Simulating Incident Management Team Response and Performance. In *Procedia Computer Science* (pp. 91–96). https://doi.org/10.1016/j.procs.2024.06.002

[4] †Apelu, D., **Macfarlane, G. S.**, Guthrie, W. S., †Adams, N. & Mazzeo, B. A. (2023). Measuring Pavement Smoothness from the Perspective of E-scooters. In *2023 Intermountain Engineering, Technology and Computing (IETC)* (pp. 114–119). https://doi.org/10.1109/IETC57902.2023.10152077

[3] **Macfarlane, G. S.** & ‡Lant, N. (2023). How Far Are We From Transportation Equity? Measuring the Effect of Wheelchair Use on Daily Activity Patterns. In C. Antoniou, F. Busch, A. Rau & M. Hariharan (Eds.), *Proceedings of the 12th International Scientific Conference on Mobility and Transport* (pp. 141–155). Springer Nature Singapore. https://doi.org/10.1007/978-981-19-8361-0_10

[2] **Macfarlane, G. S.**, Saito, M. & Schultz, G. G. (2012). Driver Perceptions at Free Right-Turn Channelized Intersections. In *Proceedings* (pp. 1128–1137). American Society of Civil Engineers. https://doi.org/10.1061/41167(398)108

[1] **Macfarlane, G. S.**, Saito, M. & Schultz, G. G. (2011). Delay Underestimation at Free Right-turn Channelized Intersections. In *Procedia - Social and Behavioral Sciences* (pp. 560–567). https://doi.org/10.1016/j.sbspro.2011.04.476

## Reports

[12] **Macfarlane, G. S.**, Schultz, G. G., Carruth, T. & Hailstone, B. (2026). *Effectiveness of Temporary Portable Rumble Strips on Work Zones*. UT-26.04, Utah Department of Transportation. https://rosap.ntl.bts.gov/view/dot/91241

[11] **Macfarlane, G. S.**, Atchley, H., Mansfield, K., Baird, T. & Gresham, C. (2024). *Activity-Based Model Implementation and Analysis Considerations*. UT-24.16, Utah Department of Transportation. https://rosap.ntl.bts.gov/view/dot/77610

[10] **Macfarlane, G. S.**, Jarvis, D., Woolley, B. & Schultz, G. G. (2024). *Simulating Incident Management Team Response and Performance*. UT-23.22, Utah Department of Transportation. https://rosap.ntl.bts.gov/view/dot/74034

[9] Guan, H., Van Hentenryck, P., Goyal, V., Hoque, J., Day, C. S., **Macfarlane, G. S.**, Erhardt, G. D. & Watkins, K. (2023). *T-SCORE Project M1-Multi-Modal Optimization: Development of Optimization Frameworks on On-Demand Multimodal Transit Systems*. M1, Transit – Serving Communities Optimally, Responsively, and Efficiently (T-SCORE) Center [UTC]. https://rosap.ntl.bts.gov/view/dot/72992

[8] **Macfarlane, G. S.** & Atchley, S. H. (2023). *Identifying Microtransit Service Areas Through Microsimulation*. UT-23.01, Utah Department of Transportation. rosap.ntl.bts.gov/view/dot/66312

[7] Schultz, G. G., Hyer, J., Holdsworth, W. H., Eggett, D. L. & **Macfarlane, G. S.** (2023). *Analysis of Benefits of UDOT's Expanded Incident Management Team Program*. UT-23.05, Utah Department of Transportation. https://doi.org/10.21949/1528563

[6] Schultz, G. G., **Macfarlane, G. S.**, Wang, B. & Davis, M. C. (2022). *Detecting Traffic Data Anomalies in Longitudinal Signal Performance Measures*. US-22.21, Utah Department of Transportation. https://rosap.ntl.bts.gov/view/dot/65833

[5] **Macfarlane, G. S.** & Lant, N. J. (2021). *Estimation and Simulation of Daily Activity Patterns for Individuals Using Wheelchairs*. UT- 21.10. https://rosap.ntl.bts.gov/view/dot/56982

[4] **Macfarlane, G. S.** & Lant, N. J. (2021). *Estimation and Simulation of Daily Activity Patterns for Individuals Using Wheelchairs*. UT-21.10, Utah Department of Transportation. https://rosap.ntl.bts.gov/view/dot/56982

[3] Schultz, G. G., **Macfarlane, G. S.**, Wang, B. & McCuen, S. (2020). *Evaluating the Quality of Signal Operations Using Signal Performance Measures*. UT-20.08, Utah Department of Transportation. https://rosap.ntl.bts.gov/view/dot/54639

[2] Zalewski, A., Sonenklar, D., Cohen, A., Kressner, J. & **Macfarlane, G.** (2019). *Public Transit Rider Origin–Destination Survey Methods and Technologies*. TCRP Synthesis 138, Transportation Research Board of the National Academies of Sciences, Engineering, and Medicine. https://doi.org/10.17226/25428

[1] Cruz, J., **Macfarlane, G.**, Xu, Y., Rodgers, M. O. & Guensler, R. (2015). *Sustainable Transportation Curricula*. National Center for Sustainable Transportation. https://escholarship.org/uc/item/3c13q43c

## Conferences & invited talks

### Conferences

P: paper/presentation, M: moderator, C: organizing committee

2026 Intermountain Engineering and Technology Conference[P], hEART European Association for Research in Transportation Annual Symposium[P], 17th International Conference on Ambient Systems Networks and Technologies (ANT)[P]

2025 Elsevier Transportation Research Symposium[P], Modeling Mobility (MoMo)[P, M], International Conference on Transportation & Development (ICTD)[P], For the Benefit of the World Conference[P]

2024 Utah Transportation Conference[P], Activity-based Modeling Symposium[P, C], International Conference on Travel Behavior Research (IATBR)[P], 15th International Conference on Ambient Systems Networks and Technologies (ANT)[P], Transportation Research Board Annual Meeting (TRB)[P]

2023 Innovations in Transportation Analysis and Planning[P, M], Move Utah Summit[P], Utah Transportation Conference[P]

2022 Mobil.TUM[P], Transportation Research Board Annual Meeting (TRB)[P]

2021 North American Regional Science Conference (NARSC)[P], Association of Collegiate Schools of Planning Annual Conference[P], Utah Transportation Conference[P]

2020 Transportation Research Board Annual Meeting (TRB)[P]

2019 Greater and Greener[P]

2018 NHTS Data for Transportation Applications Workshop[P], Innovation[P]

2017 American Public Health Association Annual Meeting[P], Transportation Planning Applications Conference[P]

2016 Innovations in Travel Modeling Conference[P, M]

2015 North American Regional Science Conference (NARSC)[P], Transportation Research Board Annual Meeting (TRB)[P]

2014 North American Regional Science Conference (NARSC)[P]

2013 North American Regional Science Conference (NARSC)[P]

### Invited talks

2025 California HSR model transferability, CalTrans Transit Research-to-Practice Symposium

2025 Sustainable Transportation, Mongolia – BYU MPA Exchange

2023 Modeling with Big Data, course presentation, Technische Universität München

2020 Using Mobile Data to Improve Transportation Sustainability, Utah Central ASCE Chapter luncheon

2020 Using Mobile Device Data to Measure Park Choice, Access, and Health, BYU Statistics Department graduate seminar

## Research funding

### External funding

### Principal investigator ($598,000 total)

2026 Macfarlane, G.S. and Brown, N.. *Building automated processes for baseline employment data*. $65,000, Utah Department of Transportation.

2026 Macfarlane, G.S. and Schultz, G.G.. *Estimating the marginal effectiveness of wrong way driver alert systems (Phase 1)*. $30,000, Utah Department of Transportation.

2025 Macfarlane, G.S. and Atchley, S.H.. *Establishing Performance Measures for Transportation and Quality of Life*. $68,000, Utah Department of Transportation.

2024 Macfarlane, G.S. and Schultz, G.G.. *Effectiveness of Temporary Portable Rumble Strips*. $60,000, Utah Department of Transportation.

2022 Macfarlane, G.S. and Brown, N.M.. *Activity-based Model Implementation and Analysis Considerations*. $70,000, Utah Department of Transportation.

2021 Macfarlane, G.S. and Schultz, G.G.. *Optimizing Traffic Incident Management Deployment in Utah*. $70,000, Utah Department of Transportation.

2021 Macfarlane, G.S., Redelfs, A.H., and Spruance, L.A.. *Equitable Access to Nutrition in Utah*. $70,000, Utah Department of Transportation.

2020 Macfarlane, G.S.. *Identifying Microtransit Service Areas through Microsimulation*. $20,000, Utah Department of Transportation.

2019 Macfarlane, G.S.. *A synthesis of passive third-party datasets used for transportation planning*. $25,000, Utah Department of Transportation.

2019 Macfarlane, G.S.. *Modeling the demand and operating characteristics for wheelchair-accessible, on-demand mobility services*. $60,000, Utah Department of Transportation.

2019 Macfarlane, G.S.. *Evaluating the Systemic Redundancy of Critical Highway Facilities*. $60,000, Utah Department of Transportation.

## Co-principal investigator ($1.44 million total ($320,000 to BYU))

2024 Watkins, K.E. (PI), Erhardt, G.D., and Macfarlane, G.S.. *The Potential for Behavioral Change in New High Speed Rail Lines*. $280,000, California High Speed Rail Authority and Deutsche Bahn.

2021 Schultz, G.G. and Macfarlane, G.S.. *Analysis of performance measures of UDOT's traffic incident management program: Phase III*. $30,000, Utah Department of Transportation.

2020 Watkins, K.E. (PI), Hunter, M.S., Van Hentenryck, P., Peeta, S., Brakewood, C., Cherry, C., Erhardt, G.D., and Macfarlane, G.S.. *T-SCORE: Transit Serving Communities Optimally, Responsibly, and Efficiently*. $1,000,000, United States Department of Transportation.

2020 Schultz, G.G. (PI) and Macfarlane, G.S.. *Evaluating Signal Performance Measures: a Longitudinal Analysis*. $70,000, Utah Department of Transportation.

2019 Schultz, G.G. (PI) and Macfarlane, G.S.. *Evaluating ramp meter delay in Utah*. $65,000, Utah Department of Transportation.

## Unfunded proposals

2023 Dashti, S. (PI), Torres-Machi, C., Mooney, M., Misra, A., Crow, D., Mallet, S., Esteghemati, M., Macfarlane, G.S., Zlatkovic, M., and Kack, D.. *Resilient, Equitable, and Sustainable Environment through Transportation (RESET)*. $3,000,000, United States Department of Transportation.

2022 Erhardt, G.D. (PI), Watkins, K.E., Brakewood, C., Pike, S.C., Macfarlane, G.S., Chavis, C., Tal, G., Hunter, M., Van Hentenryck, P., and McDonald, N.. *T-SCORE 2.0: Transit – Sustainable, Competitive, Responsive, and Equitable Center*. $2,000,000, United States Department of Transportation.

2023 Voulgaris, C.T. (PI), Forsyth, A., Pandey, V., Park, H., Handy, S., Macfarlane, G.S., Pande, A., Braun, L.M., and Noland, R.B.. *Chester: Consortium for Healthy, Equitable, and Sustainable Transportation Systems for Environmental Resilience*. $2,000,000, United States Department of Transportation.

## Internal competitive funding

## Funded research

2026 Macfarlane, G.S. and Williams, G.. *Land Use Data for Travel Forecast Modeling from Remote Sensing Data*. $25,000, Mentored Research Grant, Brigham Young University.

2024 Sowby, R.B. and Macfarlane, G.S.. *Wildfire Solutions for Western Water Utilities*. $25,000, Mentored Research Grant, Brigham Young University.

2021 Macfarlane, G.S., Guthrie, W.S., and Mazzeo, B.. *Measuring pavement smoothness from the perspective of e-scooters*. $25,000, Mentored Research Grant, Brigham Young University.

### Unfunded proposals

2020 Macfarlane, G.S., Hooley, C., Redelfs, A., and South, M.. *Using Mobile Device Data to Measure Isolation and Mental Health*. $40,000, Brigham Young University Interdisciplinary Research Grant.

## Courses

### CCE 102: Sustainable Infrastructure

Sustainability in built environments and infrastructure with a focus on global challenges and solutions. Converted from CCE 201 in Fall 2024.

### CE 361: Introduction to Transportation Engineering

Transportation systems characteristics, traffic engineering and operations, transportation planning, geometric design, pavement design, transportation safety, freight, public transport, sustainable transportation.

### CE 565: Urban Transportation Planning

Characteristics of urban transportation planning and decision making, intermodal transportation, land-use transportation interrelationships, transportation demand modeling, site impact analysis, sustainable transportation, and livable cities.

### CE 594R: Data Science for Engineers

A first-semester graduate course in programming and data science techniques: literate programming in Markdown and LaTeX, version control with git, data manipulation and visualization with R, object-oriented programming with Java.

### CE 662: Transport Simulation and Analysis

An advanced graduate course in traffic flow theory and simulation. Topics include shock wave analysis, discrete event simulation of queues and daily activity pattern choices, car following models, and traffic simulation. Laboratory assignments use MATSim and PTV Vissim simulation software.

### CE 694R: Advanced Choice Modeling

An advanced graduate course in discrete choice modeling. Theory of choice models, including estimation and validation techniques. Mode choice models for work and non-work trip purposes using multinomial and nested logit models.

## Mentoring

### Graduate committee chair (10 total, all MS)

**Connor Williams**. MS expected December 2027

**Erin Christesen**. MS expected December 2026

**Emily Youngs**, *Exploring the Link between Travel Behavior and Mental Health*. MS granted August 2024. Pursuing Ph.D. at University of Michigan

**Hayden Atchley**, *A Comparative Illustration of Trip- and Activity-Based Modeling Techniques*. MS granted August 2024

**Daniel Jarvis**, *Simulating Incident Management Team Response and Performance*. MS granted December 2023

**Natalie Gray**, *Evaluating parameter uncertainty in transportation demand models*. MS granted June 2023

**Emma Stucki**, *Evaluating equitable access to nutrition in Utah*. MS granted December 2022. Pursuing Ph.D. at Waikato University

**Gillian Riches**, *Transforming GPS points to daily activities using simultaneously optimized DBSCAN-TE parameters*. MS granted December 2022

**Christopher Day**, *Forecasting ride-hailing across multiple model frameworks*. MS granted December 2022

**Max Barnes**, *Resiliency of Utah's road network: a logit-based approach*. MS granted December 2021

**Nate Lant**, *Estimation and simulation of daily activity patterns for individuals using wheelchairs*. MS granted June 2021

## Graduate committee member (22 total, 2 current, 3 non-BYU, 3 Ph.D.)

**Kendon Moline**. MS expected December 2026

**Auzton Stutts**, *A Peer Evaluation of Safety at Signalized Intersections in Utah*. MS expected December 2026

**Cambrie Ball**, *Preparing Civil and Construction Engineers for Sustainable Practice: Systems Thinking and Interdisciplinary Learning in Undergraduate Education*. MS in Construction Management granted April 2026

**Samuel McKinnon**. Ph.D. in Mechanical Engineering; completed proposal defense

**Adam W. Hill**, *Using unmanned aerial vehicles to facilitate traffic incident management*. MS granted April 2024

**Ian MacGregor**, *Implementing the safe system approach at intersections in Utah*. MS granted December 2024

**Sam Runyan**, *Effect of roadway lighting on safety in Utah*. MS granted December 2024

**Joel Hyer**, *Analysis of benefits of UDOT's expanded incident management team program*. MS granted April 2024

**Matthew Davis**, *Effectiveness of intelligent transportation systems on Utah roadways*. MS granted December 2023

**William Charlton**, *Web-based data visualization in support of agent-based microsimulation models*. Ph.D. granted October 2023. Technische Universität Berlin

**Wang Bangyu (Bruce)**, *Evaluating and advancing automated traffic signal performance measures: Statistical and machine learning approaches*. Ph.D. granted August 2023

**Tomas Barriga**, *Using severity weighted risk scores to prioritize safety funding in Utah*. MS granted August 2023

**Benjamin Meek**, *Load-deformation behavior of tension-only X-brace roof truss diaphragms*. MS granted April 2023

**Mylan Cook**, *Physics-guided modeling of acoustic environments using limited spatio-spectro-temporal data*. Ph.D. in Physics granted June 2023

**Cory Ward**, *An evaluation of the safe speed limit setting procedure and tool for Utah (Project)*. MS granted December 2022

**Samantha Lau**, *Analysis of using V2X DSRC equipped snowplows to request signal preemption*. MS granted August 2022

**Tanner Daines**, *Evaluating ramp meter delay in Utah*. MS granted April 2022

**Logan Bennett**, *Analysis of benefits of an expansion to UDOT's incident management program*. MS granted August 2021

**Camille Lunt**, *Crash analysis methodology for segments of Utah highway*. MS granted April 2021

**Chad Vickery**, *Quantifying the conditioning period for geogrid-reinforced aggregate base materials through cyclic loading*. MS granted August 2020

**Michael Sheffield**, *Impacts of changing the transit signal priority requesting threshold on bus performance and general traffic: a sensitivity analysis*. MS granted June 2020

**Michael Adamson**, *An analysis of decision boundaries for left-turn treatments*. MS granted April 2019

**Nico Boyd**, *Accessibility to urban parks and health outcomes on the neighborhood level*. MS granted August 2018. Georgia Institute of Technology

**Zhang Bingling**, *Friendship influences on air travel: a social autoregressive analysis*. MS granted August 2014. Georgia Institute of Technology

## Undergraduate funded research (28 total)

2026–present **Christian Jarvis**, travel behavior

2026–present **Elise Lefler**, travel behavior

2026-present **Elizabeth Woodford**, demand microsimulation

2025–2026 **Kaylen Turley**, demand microsimulation

2024–present **Tyler Carruth**, work zone safety

2024–2026 **Benjamin Hailstone**, work zone safety

2024–2026 **Connor Williams**, demand microsimulation

2024–2025 **Kaleigh Squires**, demand microsimulation

2023–2024 **Kamryn Mansfield**, demand microsimulation. Ph.D. student at University of Tennessee

2022–2024 **Brynn Woolley**, demand microsimulation. Ph.D. student at University of Michigan

2022 **Jeremy Raine**, community resources

2021–2022 **Harrison Holdsworth**, demand microsimulation

2022 **Jonathan Orton**, e-scooters and pavements

2021–2023 **Dylan Apelu**, e-scooters and pavements. Completed MS at Georgia Institute of Technology

2021–2022 **Matthew Davis**, signal performance data. Jointly mentored with Grant Schultz. Ph.D. student at University of Tennessee

2020–2022 **Hayden Atchley**, demand microsimulation. Completed MS at BYU

2021–2022 **Nicole Adams**, e-scooters and pavements

2021–2022 **Liv Neeley**, e-scooters and pavements

2020–2022 **Kaeli Monahan**, community resources and passive data

2020–2022 **Shannon Anderson**, V2X data. Jointly mentored with Grant Schultz

2020–2021 **Corey Ward**, ramp meter evaluation. Jointly mentored with Grant Schultz. Completed MS at BYU

2018–2021 **Michael Copley**, third-party passive data. Completed MS at University of Illinois

- 2020–2021 **James Umphress**, ramp meters. Jointly mentored with Grant Schultz. Completed MS at Oregon State University
- 2020–2021 **Christopher Day**, demand microsimulation. Completed MS at BYU
- 2020–2021 **Emma Stucki**, community resources. Completed MS at BYU
- 2020–2021 **Gillian Martin Riches**, community resources. Completed MS at BYU
- 2019–2021 **Natalie Gray**, network resiliency. Completed MS at BYU
- 2019–2020 **Max Barnes**, network resiliency. Completed MS at BYU
- 2020–2021 **Kim Munseok**, demand microsimulation
- 2018–2019 **Christian Hunter**, demand microsimulation. Completed MS at University of Texas at Austin
- 2019 **Christian Vanderhoeven**, demand microsimulation. Completed MS at University of Washington
- 2019–2020 **Hayden Anderson**, e-scooters. Completed MS at University of California
- 2019 **Emily Andrus**, signal performance data. Jointly mentored with Grant Schultz
- 2019–2020 **Sabrina McCuen**, signal performance data. Jointly mentored with Grant Schultz

## Department honors coordinator (3 students)

- 2024 **Becca Apgar**, *Development and demonstration of an apparatus for assessing frost-heave susceptibility of soil*
- 2023 **Daria Sofia Velasco-Vega**, *Thermal performance of thin-shell concrete dome structures*. Pursuing Ph.D. at Purdue University
- 2022 **Emma Kratz-Bailey**, *Accessible methods, novel arrangement: Developing self-centering composite structural frames for highly resilient buildings*

## Other mentoring

- 2024 **Grace Gibson**, *Gendered Journeys: analyzing the context, barriers, and solutions for women in public transit*. Mentor for BYU Global Women's Studies Minor capstone project

## Civil engineering capstone team mentor (26 students)

- 2024–2025 **Land development capacity system design**. Sponsored by UDOT. Students: George Cicotte, Emily Jacobsen, Tacoma Parkinson.
- 2023–2024 **BYU household travel survey**. Sponsored by BYU Sustainability Office. Students: Megan Hungerford, Ellie Johns, Kamryn Mansfield, Myrranda Salmon.
- 2022–2023 **SR-140 Corridor alternatives**. Sponsored by Bluffdale City. Students: Clinton Childers, Robert Mickelson, Trevor Mickelson, Joseph Wells.
- 2021–2022 **BYU household travel survey**. Sponsored by BYU Sustainability Office. Students: Nicole Adams, Hayden Atchley, Kyle Leatham, Daniel Jarvis.
- 2020–2021 **Forecasting demand for future FrontRunner scenarios**. Sponsored by Utah Transit Authority. Students: Gillian Martin Riches, Tomas Barriga, Landon Pratt, Cole Larsen.
- 2019–2020 **UTA microtransit pilot evaluation**. Sponsored by Utah Transit Authority. Students: Christian Hunter, Austin Martinez, Elizabeth Smith.
- 2018–2019 **Demand for wheelchair-accessible vehicles**. Sponsored by Utah Transit Authority. Students: Nate Lant, Byron Yates, Cody Irons, Matthew Strong.

## Awards & honors

2023 **TUM Global Visiting Professor**. Selected to enrich the research culture at Technische Universität München through innovative approaches and exploration of new, cutting-edge research fields.

2022 **Most Influential Faculty**. Given to the Civil Engineering faculty member whom graduating seniors named as the most influential on their undergraduate education.

2022 **ASCE ExCEEd Teaching Fellow**. Participated in a week-long intensive teacher development program.

2011–2013 **Dwight David Eisenhower Graduate Fellowship**. Full doctoral funding from the Federal Highway Administration; one of five awards nationally. Awarded a supplemental grant in 2013.

2012 **Eno Center for Transportation Leadership Development Conference**. Participated in the program; nominated by the Ivan Allen Jr. College of Liberal Arts at Georgia Tech.

## Citizenship & service

### External citizenship

2023–present **NCHRP 08-184**: Peer review panel member. Framework for Assessing Induced Demand Effects of Various Roadway Investments

2019–2025 **Transportation Research Board, AEP50: Travel Demand Forecasting**: Committee member; chair, travel forecasting resources subcommittee; editor of tfresource.org

2014–2022 **Transportation Research Board, AMS20: Economics and Land Development**: Committee member

2019–2021 **Transportation Research Board Young Members Council**: Planning and Environment subcommittee chair

2022–present **Provo City Transportation and Mobility Advisory Commission**: Member, appointed by Mayor Michelle Kaufusi

### Journal reviewer

Transportation Research Part A: Policy and Practice, Smart Cities, Transportation Research Record, Environment and Planning B: Urban Analytics and City Science, International Journal of Sustainable Transport, Journal of Public Transportation

### Professional memberships

2022–present American Society of Civil Engineers

2020–2022 Zephyr Foundation

2009–2013; 2018–2020 Institute of Transportation Engineers

2009 Tau Beta Pi. Utah β ’09

2013–2018 Young Professionals in Transportation. Organizing co-chair of Triangle NC chapter

2011–2013 American Public Transportation Association scholar task force

## Internal citizenship

2023–present **Department undergraduate committee**: Co-chair. Member since 2021. Lead curriculum revisions for the civil engineering program and ABET accreditation and continuous-improvement efforts.

2023–present **Office of Civic Engagement**: Faculty advisory committee member. Participate in decisions related to the civic engagement minor and advise on campus activities.

2019–present **Department honors program**: Coordinator. Encourage students to participate in the honors program and serve on departmental honors thesis committees.

2018–2021 **Department faculty development and capital improvement committee**: Member